Energy-Secure, Carbon-Aware, and Formally Verifiable Autonomous Networks for Canadian 6G

Thesis Contributions Relevant to Bell Mobility's Strategic Priorities

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Outline

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2. Contribution 1 — CALASH: Carbon-Aware Lifecycle Routing
3. Contribution 2 — CASTER-ZT: Zero-Trust Autonomous Recovery
4. Contribution 3 — GAPF: TinyML Autonomous Routing
5. Evidence Alignment & Positioning
6. Collaboration Proposal & Ask

Bell Is Already Building the 6G Foundation

Bell's Confirmed 6G-Adjacent Investments

- **AI-native RAN** — Ericsson live-network test
  $\uparrow$20% DL throughput, $\uparrow$10% spectral efficiency (Apr 2025)
- **Cloud/Open RAN** — Nokia deal:
  Red Hat OpenShift + AirScale Open Fronthaul (MWC 2025)
- **5G+ Advanced** — SA core, 3800 MHz, "paving the way towards 6G" (BCE 2026)
- **Bell AI Fabric** — sovereign hydro-powered edge compute
- **NTN / Direct-to-Cell** — AST SpaceMobile partnership

What Is **Not Yet** Public

- No Bell-authored 6G white paper
- No near-RT RIC commercial deployment (TELUS leads)
- No carbon-aware routing framework
- No zero-trust O-RAN architecture published
- No first-party 6G patent filings (35 total Bell patents)

The Strategic Frame

"The question is not when 6G arrives — it is how to make the

Three Open Challenges Bell Faces Today

#1 — Carbon & Energy Gap

- Scope 3 = **91%** of BCE footprint
- Purchased goods & services = **59%** of total — *excluded* from reduction target
- No disclosed carbon-aware routing or feature-level AI/RAN carbon attribution
- 2030 target: −58% Scope 1+2, −42% Scope

#2 — Autonomous RAN Security

- O-RAN WG11: **160+** distinct threats identified
- New AI/ML threat classes: model poisoning, inversion, stealing
- Bill C-8: critical infrastructure cyber obligations tightening
- No public Bell zero-trust O-RAN framework

O-RAN Alliance WG11 2025/2026

#3 — Rural / Indigenous IoT

- Only **65.7%** of First Nations reserves have 50/10 Mbps coverage
- CRTC 2025: Indigenous consent required for funded builds
- No Bell TinyML/sub-1W program confirmed
- Northern deployment economics vs. NTN coverage

CALASH — Carbon-Aware Lifecycle-Adaptive Self-Healing Routing

The Core Insight

Embodied carbon of one IoT/RAN node $\approx$**10 000 gCO_2eq**
Operational carbon per packet $\approx$**0.01 gCO_2eq**

$\Rightarrow$ **Ratio 10^6:1** — lifetime extension *is* the carbon strategy.

Four integrated pillars:

1. **CADR** — Adaptive compressive sensing ($\rho \in [0.3, 0.8]$), fidelity $F > 0.99$
2. **CARE** — Lyapunov-DQN carbon-aware router (grid intensity-aware)
3. **SHDR** — MAPE-K self-healing (failure

Formal Guarantee — CDL Pareto Bound

For any control parameter $V > 0$, simultaneously:

$$\bar{D} \geq \bar{D}^* - \frac{B}{V} \quad \text{and} \quad \bar{C}_{\text{tot}} \leq \bar{c} + \frac{B}{V}$$

Delivery $\bar{D}$ stays near-optimal *and* carbon $\bar{C}_{\text{tot}}$ stays bounded — *simultaneously*.

Metric	CALASH	Best Baseline
Network lifetime	+60%	0%
Lifecycle carbon (LCI)	−33%	−8%

CASTER-ZT — Zero-Trust Shielded Recovery for Autonomous RAN

The O-RAN Security Problem

Open RAN expands the attack surface: open interfaces, multi-vendor xApps, AI/ML pipelines, O-Cloud. O-RAN WG11 catalogued **160+ distinct threats** including model poisoning, membership inference, and AI supply-chain attacks.

Near-RT RIC budget: **10–1000 ms** — any security guardrail must stay inside.

CASTER-ZT 5-stage pipeline:

1. **GNN encoder** — topology-aware state representation
2. **Learned recovery policy** — GNN guided

Key Results

Metric	
Decision latency	**3.07 ms** $<$ 10
Rogue detection (12 cells)	**74.2%**
Rogue detection (100 cells)	**98.3%**
Recovery score ω_{rec}	**0.733**
Formal properties proved	

Canadian Regulatory Hook

- **Bill C-8**: critical-infrastructure cyber obligations
- **CSE Telecoms Cyber Resilience**

GAPF — TinyML Graph-Attentive Policy Fusion for Rural/IoT Networks

The Problem: Sub-1W Autonomy at the Edge

Canada has **65.7%** broadband coverage on First Nations reserves.
Bell's public-safety and IoT portfolio requires autonomous, battery-powered sensors for **remote monitoring, NG9-1-1, wildfire, and flood sensing**.

Existing MARL controllers are too large for constrained hardware (>10k parameters).

Performance vs. Baselines

Metric	GAPF	Best Bas
Parameters	**1,473**	19,20
Memory reduction	**13**×	—
Energy reduction	**2–3.5**×	—
PDR	≥**98%**	91%
TinyML (Cortex-M4)	✓	×

GAPF bi-level architecture:

- **Inner loop** — MARL (QMIX/QTRAN) with monotonic $Q \cdot K$ attention scalarization (formally Pareto-guaranteed optimal

Bell Application Framing

- **Rural/Indigenous sensor mesh**: reduce truck rolls
- **Bell AI Fabric edge inference**:

Thesis vs. Bell's Confirmed Strategic Priorities

Bell Priority	CALASH	CASTER-ZT	GAPF
AI-native RAN efficiency	Carbon-aware Lyapunov-DQN routing	GNN-based recovery policy	MARL meta-controller (1,473 params)
Cloud / O-RAN path	CDL Pareto bound for SLA + carbon	ZT shield for xApp/rApp security	Distributed edge control
Network energy (BCE −58% Scope 1+2)	**Addresses embodied + operational**	–	Reduces active-transmit energy 2–3.5×
Scope 3 embodied carbon (91% of footprint)	**ISO 14040 lifecycle engine**	–	Lifetime extension ⇒ deferred hardware
O-RAN / near-RT RIC (10 ms budget)	–	**3.07 ms, 0% FP, 10 proved props**	–
Bill C-8 cyber compliance	–	**Formally verifiable guardrail**	–
Rural / Indigenous connectivity	+ lifetime ⇒ fewer replacements	+ resilient recovery	**TinyML Cortex-M4, 98% PDR**
Bell AI Fabric / MEC	Grid-carbon-aware edge of	Secure xApp deployment	Sub-1W distributed infer

Collaboration Proposal — Three Actionable Pathways

Pathway A — MITACS

CALASH on Bell IoT/RAN testbed

- Duration: 6–12 months
- Goal: Validate carbon-aware routing on real NB-IoT/LTE-M traffic and provincial grid-carbon data
- Bell teams: Network Operations, ESG, AI Fabric
- Deliverable:

Pathway B — NSERC CRD

CASTER-ZT O-RAN Security Lab

- Duration: 2–3 years
- Goal: Hardware-in-the-loop near-RT RIC zero-trust testbed, rogue xApp emulation
- Bell teams: Bell Cyber, Mobility RAN Automation, PSBN
- Deliverable: Open-source

Pathway C — Bell AI Fabric Pilot

GAPF TinyML Edge Deployment

- Duration: 6 months PoC
- Goal: Deploy GAPF on Cortex-M4 nodes in a rural/Indigenous sensor mesh, measure truck-roll reduction
- Bell teams: Bell IoT, Public Safety, AI Fabric
- Deliverable: TinyML

Closing Statement

Bell can lead Canadian 6G — not by announcing 6G first,
but by proving that autonomous networks can be
carbon-aware, formally verifiable, and sub-1W at the edge.

Under Canadian geography, regulation, and climate constraints.

Publications

IEEE TGCN (Scopus Q1)
GAPF + CALASH accepted
IEEE TMC (under review)
CASTER-ZT

Code & Reproducibility

Open-source repositories
All experiments reproducible
Python + PyTorch + formal proofs
Available for Bell review

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Thank you — questions welcome

Backup — CALASH: CDL Theorem (Proof Sketch)

Virtual Queue $Z(t)$ and Lyapunov Drift-Plus-Penalty

Define virtual carbon queue: $Z(t+1) = \max\left[Z(t) + C_t - \bar{c},\ 0\right]$
Lyapunov function: $L(t) = \frac{1}{2}Z(t)^2$

Minimize per-slot drift-plus-penalty: $\Delta L(t) + V \cdot C_t^{\text{total}} - V \cdot \lambda \cdot D_t$

Under standard Lyapunov stability analysis, the time-average:

$$\bar{D} \geq \bar{D}^* - \frac{B}{V} \quad \text{and} \quad \bar{C}_{\text{tot}} \leq \bar{c} + \frac{B}{V}$$

hold simultaneously for any $V > 0$, where B is a finite constant.

Interpretation: Increasing V trades smaller carbon slack for better delivery optimality. The Pareto front is convex and fully traceable by varying a single scalar.

Backup — CASTER-ZT: 10 Proved Formal Properties

1. **Safety:** No unsafe action admitted by the shield
2. **Liveness:** Recovery is always eventually initiated
3. **Bounded admissibility:** Only actions within policy envelope admitted
4. **Rogue isolation:** Rogue agents cannot influence quorum decisions
5. **False-positive bound:** FP rate $\leq \delta$ (conformal coverage)
6. **Timing compliance:** Decision latency $<$ 10 ms budget
7. **Conformal coverage:** Distribution-free coverage guarantee
8. **Calibration convergence:** MC-Dropout calibration error $\to 0$
9. **CDL defense-in-depth:** 5 layers (detect / scope-reduce / defer / escalate / block)
10. **Model integrity:** GNN encoder invariant to permutation and node dropout

Proof method: Formal induction on the shield transition system + conformal prediction theory + Lyapunov stability for recovery convergence.

Backup — Canadian Carrier Comparison (6G-Adjacent)

Dimension	Bell Mobility	Rogers	TELUS
AI-native RAN	Strongest: Ericsson live field test (+20% DL, Apr 2025)	5G SA + RedCap, less AI-RAN differentiation	AI-RIC via Samsung CognitiV NOS
O-RAN / RIC maturity	Cloud RAN (Nokia), no public near-RT RIC	Cloud RAN trial (live event), no commercial O-RAN	Canada's 1st commercial vRAN/O-RAN + AI-RIC (Samsung)
MEC / edge	Leads: AWS Wavelength + Google DCE + AI Fabric	Less public MEC evidence	Sovereign AI factory, strong but Bell leads MEC
Sustainability	Strong ESG targets, *no carbon-aware routing*	CTDA participant	Strong brand, Bell stronger on AI-RAN energy
Security / ZT	Strong cyber business, *no public O-RAN ZTA*	Post-outage resiliency focus	Signs Global Coalition on 6G Security; stronger O-RAN security visibility
6G PHY research	Cohere/OTFS via Bell Ventures, NTN/AST	Limited public 6G PHY work	Limited public 6G PHY work

Sources: Ericsson press release Apr 2025; Nokia MWC 2025; Samsung/TELUS announcements 2024–2025; BCE 2024 Annual Financial Report.

Backup — What Bell Would Need Before Collaboration

Requirement	CALASH	CASTER-ZT	GAPF
Reproducible code + data	✓Available	✓Available	✓Available
Hardware-in-loop demo	Sensor/edge device energy test	O-RAN near-RT security testbed	Cortex-M4 IoT node test
Telecom scenario mapping	Provincial grid-carbon + NB-IoT/LTE-M SLA	E2/A1/O1/R1 threat model, rogue xApp emulation	LTE-M / NB-IoT / mMTC traffic
Baseline comparisons	Always-active, always-passive, energy-only routing	ZTA detectors, rule-based recovery	RPL, LEACH, MARL/GNN, TinyML
Formal due diligence	ISO 14040 assumptions, LCA database	10 formal proofs, adversarial ML stress-test	Energy + latency on real hardware
Deployment pathway	MITACS Accelerate	NSERC CRD	Bell Innovation Fund / NSERC Alliance

Current status: Code available, thesis chapters in final review, formal proofs complete. Ready for a structured Bell technical review session.